Journey: resolving an OSA decline

Six stages from spotting a drop to confirming recovery. Yellow notes are AI opportunities, scored in frame 3.

	1 Detect	2 Scope	3 Diagnose	4 Prioritise	5 Act	6 Track
Doing	Reviews weekly OSA dashboard and spots a drop	Drills from retailer to region to store to SKU	Cross-checks POS, store stock, DC stock, shipments and audits	Decides which stores and SKUs to fix first	Briefs field reps, raises with supply chain or retailer	Checks later weeks to see if OSA recovered
Pain	Drops seen late; noise vs real decline unclear	Reports live in different tools and grains	Data shows what happened, not why. Needs analyst help	Gut feel; no consistent lost-sales view	Wrong owner contacted; actions not standardised	No link between action taken and result
Data used	Retailer OSA feed	OSA plus store and SKU master	POS, store and DC inventory, shipments, audits, promo calendar	POS baseline, price, OSA	Contacts, action playbook	OSA and POS after the action
Feeling	Concerned	Overloaded	Frustrated	Uncertain	Pressured	Unsure it worked
AI could help	**O1** Flag real OSA drops **O8** Predict OOS early	**O2** Ask in plain language **O10** Match SKU and store codes	**O3** Rank causes with evidence **O9** Read shelf photos	**O4** Lost-sales ranked worklist	**O5** Next-best action and owner **O6** Draft messages **O7** Weekly summary	**O11** Measure action effectiveness

Biggest time sink is stage 3, Diagnose. That is where the copilot's core AI sits; stages 1 and 4 support it with deterministic rules and maths.

Problem and user

The root cause from Step 1 that this map designs against.

Root-cause problem
Retail Execution and Sales Ops Managers cannot quickly explain why On-Shelf Availability is falling, because retailer, POS, inventory and store-execution data sit in separate systems. Investigation is manual, causes stay unclear, and fixes are not ranked by sales impact.

Primary user
Retail Execution / Sales Ops Manager. Owns OSA for a retailer or region, reviews weekly dashboards, coordinates reps and supply chain.

Second stakeholder
Supply Chain / Demand Planner. Receives escalations when the cause is upstream, such as a DC shortage or order cuts.

Job to be done
OSA dropped. Which store-SKUs matter most, why did it drop, and who needs to do what this week?

Proposed solution
An AI copilot over unified OSA, POS, inventory and store-execution data. It flags at-risk stores, explains likely causes with evidence, and ranks corrective actions by estimated sales impact. The manager stays in control of every action.

Scope boundaries

What the MVP copilot does, never does, and leaves out of this capstone.

Should do	Should NOT do	Out of scope
- Flag store-SKUs where OSA is below target or dropped vs a 4-week baseline - Rank the top 1 to 3 likely causes from a fixed list, each with confidence and the signal values behind it - Estimate lost sales with a visible formula and rank the worklist by it - Recommend the next action and owner from an approved playbook - Answer plain-language questions about the OSA data and show the source - Say insufficient evidence when signals are missing or conflict - Record the manager's accept or reject on each diagnosis	- Send messages, place orders or change replenishment on its own - State one definitive cause without evidence - Generate or alter numbers; all figures come from the data layer - Invent actions outside the playbook - Judge or rank individual field reps or store staff - Show one retailer's data to users scoped to another - Answer off-topic questions or act as a general chatbot - Replace physical store audits	- Live integrations and ETL from retailer portals, ERP or POS - Master-data cleanup and SKU or store code matching (O10) - Forecasting, shelf-photo vision and closed-loop learning (O8, O9, O11) - Pricing and promotion decisions - Ticketing or CRM workflow

Prototype data: synthetic and pre-joined. Suggested size: 1 retailer, about 50 stores, about 20 OTC SKUs, 12 weeks.

Impact vs feasibility

Each opportunity scored 1 to 5 on both axes. High means a score of 4 or 5. Priority = Impact x Feasibility.

Roadmap: high impact, hard to build now

O8 Predict OOS early I5 F2

O11 Action effectiveness I4 F2

O9 Shelf-photo vision I4 F1

Build now: high impact, feasible

O4 Lost-sales worklist I5 F5

O3 Root-cause diagnosis I5 F4

O1 At-risk detection I4 F5

O5 Next-best action I4 F4

Deprioritise: lower impact, hard

O10 Match SKU and store codes I3 F2

Add-ons: lower impact, easy

O2 Plain-language Q&A I3 F4

O6 Draft messages I3 F5

O7 Weekly summary I2 F5

	Opportunity	ID	AI type	Impact	Feasibility	Priority	Decision
1	Lost-sales impact and ranked worklist	O4	Deterministic calc, LLM explains		5	5	25	MVP
2	Root-cause diagnosis with cited evidence	O3	Hybrid: rule checks plus LLM reasoning		5	4	20	MVP
3	At-risk store-SKU detection	O1	Rules and statistics, no LLM		4	5	20	MVP
4	Next-best action and owner	O5	Retrieval over action playbook		4	4	16	MVP
5	Draft messages to reps and planners	O6	LLM generation		3	5	15	Stretch
6	Ask questions in plain language	O2	LLM tool-calling over fixed metrics		3	4	12	MVP interface
7	Weekly executive summary	O7	LLM summarisation		2	5	10	Not now
8	Predict OOS before it happens	O8	ML forecasting		5	2	10	Roadmap
9	Closed-loop action effectiveness	O11	Causal or uplift analysis		4	2	8	Roadmap
10	Match SKU and store codes across sources	O10	Entity resolution		3	2	6	Out of scope
11	Shelf-photo recognition	O9	Computer vision		4	1	4	Roadmap

Key call

Detection (O1) and lost-sales maths (O4) stay deterministic, so the LLM never computes a number. AI is used where judgment is needed: weighing conflicting signals into a ranked cause (O3), matching the cause to an action (O5), and answering questions (O2). O2 scores lower alone but is kept as the interface to the MVP.

Impact: how much it cuts investigation time or recovers lost sales. **Feasibility:** buildable and testable in the capstone with synthetic data, on a free-tier deploy, at acceptable risk.

Why not just ChatGPT?

Four things a single unconstrained prompt cannot give the manager.

Grounded in joined data
Answers come from one store-SKU-week dataset across OSA, POS, inventory and execution, not from what the user pastes in.

Numbers are computed
Detection and lost-sales maths are deterministic and auditable. The LLM explains and reasons; it never calculates.

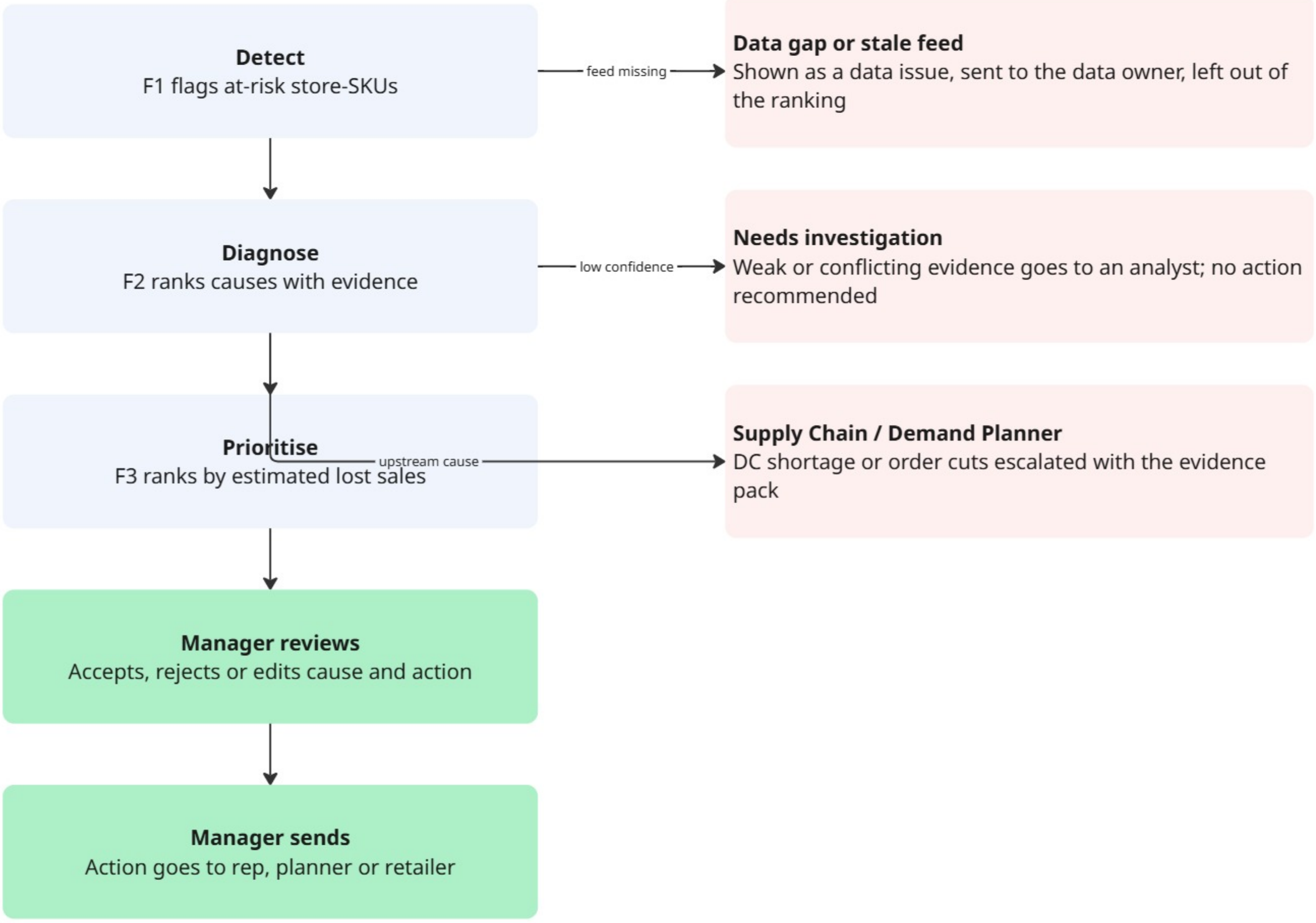
Constrained reasoning
A fixed cause list, evidence citations and an approved playbook keep outputs consistent across managers and regions.

Defined human control
Manager approval before any action, escalation paths for low confidence and upstream causes, and an audit log.

In one line
ChatGPT can talk about OSA. This copilot reasons over your joined data, inside fixed rules, and hands every decision back to the manager.

Human in the loop

Green steps are the manager's decisions. Red boxes are escalation paths.



Every accept or reject is logged with a reason. That log becomes the measure of diagnosis accuracy, a supporting metric for the Step 3 PRD.

Feature, AI capability and guardrails

Each MVP feature mapped to the capability behind it, what grounds it, and the guardrails it must respect.

	Feature	AI capability	Grounded in	Output	Guardrails
1	F1 At-risk detection (O1)	Rules and statistics: OSA below target, or a drop of more than X points vs the 4-week baseline	OSA by store-SKU-week	List of flagged store-SKUs	Thresholds visible and configurable. Minimum weeks of history before flagging. Missing feeds flagged as a data gap, not as out-of-stock.
2	F2 Root-cause diagnosis (O3)	Hybrid: rule checks build an evidence pack, the LLM weighs it and ranks causes	POS units, store on-hand, DC stock, shipments, audit results, promo flag	Top 1 to 3 causes with confidence and evidence lines	Fixed cause list: DC shortage, store ordering miss, phantom inventory, shelf execution gap, demand or promo spike, delisting, data gap. Each cause cites at least 2 signals with values. Low confidence or conflict becomes Needs investigation. Output schema validated, low temperature.
3	F3 Impact prioritisation (O4)	Deterministic calculation; the LLM only explains the ranking	Baseline weekly units x price x weeks affected x (1 - OSA)	Ranked worklist with estimated lost sales	Formula and inputs shown. LLM text checked against computed values before display. Labelled as an estimate.
4	F4 Next-best action (O5)	Retrieval over an approved playbook: cause, action, owner, target turnaround	Action playbook plus the diagnosed cause	Recommended action and owner, with the playbook entry cited	Only playbook actions. Cites the entry used. Manager approves before anything leaves the tool. No auto-execution.
5	F5 Plain-language questions (O2)	LLM tool-calling over predefined metrics and filters	Unified OSA dataset, read-only	Answer with a table or chart and the query used	Read-only. Predefined metrics only. Refuses off-topic questions. Retailer-level access scoping. Says when the data cannot answer.
6	F6 Draft message (O6, stretch)	LLM generation from templates	Approved action plus store-SKU facts	Editable draft to a rep or planner	Draft only; a human sends it. No cross-retailer data. Figures inserted from data, never generated.

Cause list used by F2
Each cause has a signal pattern the rule checks look for.

DC shortage: DC stock low, shipments cut

Store ordering miss: DC fine, store on-hand low, no recent order

Phantom inventory: system on-hand above zero, zero POS for 2+ weeks

Shelf execution gap: stock in store, audit shows not on shelf

Demand or promo spike: POS well above baseline, promo flag on

Delisting: item removed from the planogram

Data gap: feed missing or stale